

1. Assume there are only two goods, X and Y .
Prove that quasi-concave utility $U(X, Y)$ is identical to $dMRS/dX < 0$.

Answer:

For a twice differentiable function $U(X, Y)$ to be quasi-concave, its Hessian matrix must satisfy condition involving the Bordered Hessian determinant, which is used in the context of optimizing quasi-concave functions

The Hessian matrix of $U(X, Y)$ is given by-

$$H = \begin{bmatrix} U_{XX} & U_{XY} \\ U_{XY} & U_{YY} \end{bmatrix}$$

For a quasi-concave function, a necessary condition is that the determinant of the bordered Hessian matrix, with the first-order

partial derivatives of $U(x, Y)$ as the borders must be non-positive. The Bordered Hessian is -

$$B_H = \begin{bmatrix} 0 & U_x & U_Y \\ U_x & U_{xx} & U_{xY} \\ U_Y & U_{xY} & U_{YY} \end{bmatrix}$$

According to condition $\det(B_H) \leq 0$

$$\text{Now, } \det(B_H) = 0 \cdot (U_{xx} \cdot U_{YY} - U_{xY} \cdot U_{xY})$$

$$- U_x (U_x \cdot U_{YY} - U_Y \cdot U_{xY}) + U_Y (U_x \cdot U_{xY} - U_Y \cdot U_{xx})$$

$$\text{or, } \det(B_H) = 0 - U_x^2 \cdot U_{YY} + U_x \cdot U_Y \cdot U_{xY} + U_x \cdot U_Y \cdot U_{xY} - U_Y^2 \cdot U_{xx}$$

$$= -U_x^2 U_{YY} + 2U_{xY} U_x U_Y - U_Y^2 U_{xx}$$

Now, according to condition,

$$-U_x^2 U_{YY} + 2U_{xY} U_x U_Y - U_Y^2 U_{xx} \leq 0$$

$$\therefore U_{xx} U_Y^2 - 2U_{xY} U_x U_Y + U_{YY} U_x^2 \leq 0$$

If $U(x, y)$ is strictly-concave, then the inequality becomes strict. That is,

$$U_{xx} \cdot U_y^2 - 2U_{xy} U_x U_y + U_{yy} U_x^2 < 0$$

Now, for usual convex indifference curve with utility function $U(x, y)$ -

$$MRS = \frac{U_x}{U_y}$$

$$\& \frac{\partial MRS}{\partial x} = \frac{d}{dx} \left(\frac{U_x}{U_y} \right)$$

$$\therefore \frac{\partial MRS}{\partial x} = \frac{U_{xx} U_y - U_x U_{xy}}{U_y^2} \quad \left[\text{Using } \frac{u}{v} \text{ rule} \right]$$

Now, for $\frac{\partial MRS}{\partial x} < 0$, the numerator must be negative. That is,

$$U_{xx} U_y - U_x U_{xy} < 0$$

Thus, Quasi-concave utility $U(x, y)$ is identical to $\frac{\partial MRS}{\partial x} < 0$.

2. $U = XY^3$. Find the demands for X and Y .

Answer:

$$\text{Here, Max}_{X,Y} U = XY^3$$

$$\text{s.t. } I = P_X X + P_Y Y$$

$$\therefore \text{Max}_{X,Y,\lambda} L = XY^3 + \lambda [I - P_X X - P_Y Y]$$

$$\frac{\partial L}{\partial X} = Y^3 - \lambda P_X = 0 \quad \text{or, } Y^3 = \lambda P_X \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial Y} = 3XY^2 - \lambda P_Y = 0 \quad \text{or, } 3XY^2 = \lambda P_Y \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = I - P_X X - P_Y Y = 0 \quad \text{or, } I = P_X X + P_Y Y \quad \text{--- (iii)}$$

From (i) and (ii),

$$\frac{Y^3}{3XY^2} = \frac{\lambda P_X}{\lambda P_Y}$$

$$\text{or, } Y^* = \frac{3P_X X}{P_Y} \quad \text{--- (iv)}$$

Putting value of Y in (iii),

$$I = P_X X + P_Y \cdot \frac{3P_X X}{P_Y}$$

$$\text{or, } I = P_x X + 3 P_x X$$

$$\text{or, } I = 4 P_x X$$

$$\therefore X^* = \frac{I}{4 P_x}$$

So, Marshallian Demand Functions are-

$$X^* = \frac{I}{4 P_x}$$

& From (iv),

$$Y^* = \frac{3 P_x}{P_y} \cdot \frac{I}{4 P_x}$$

$$\therefore Y^* = \frac{3I}{4 P_y}$$

3. $U = X + 2Y$; $P_x = 2$; $P_Y = 5$; $I = 50$. How much of X and Y should you buy?

Answer:

Here,

Demands $X^* = \frac{I}{P_x}$ if $P_x < P_Y$ or, $X = 0$ if $P_x > P_Y$

$Y^* = \frac{I}{P_Y}$ if $P_x > P_Y$ or, $Y = 0$ if $P_x < P_Y$

Now, Max $U = X + 2Y$
 x, Y

s.t. $50 = 2X + 5Y$

So, $L_{x,Y,\lambda} = X + 2Y + \lambda [50 - 2X - 5Y]$

$$\frac{\partial L}{\partial X} = 1 - 2\lambda = 0 \quad \text{or,} \quad 1 = 2\lambda \quad \text{--- (i)}$$

$$\frac{\partial L}{\partial Y} = 2 - 5\lambda = 0 \quad \text{or,} \quad 2 = 5\lambda \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial \lambda} = 50 - 2X - 5Y = 0 \quad \text{--- (iii)}$$

Dividing (i) by (ii),

$$\frac{1}{2} = \frac{2\lambda}{5\lambda}$$

or, $\frac{1}{2} \neq \frac{2}{5}$; in this case we get corner solution.

$$\text{Here, } MRS = \frac{MU_x}{MU_y} = \frac{1}{2} ; \frac{P_x}{P_y} = \frac{2}{5}$$

$$MRS > \frac{P_x}{P_y}$$

That is, Budget line is flatter than Indifference Curve as shown in figure.

Now,

$$\frac{MU_x}{P_x} = \frac{1}{2}$$

$$\frac{MU_y}{P_y} = \frac{2}{5}$$

$$\text{Thus, } \frac{MU_x}{P_x} > \frac{MU_y}{P_y}$$

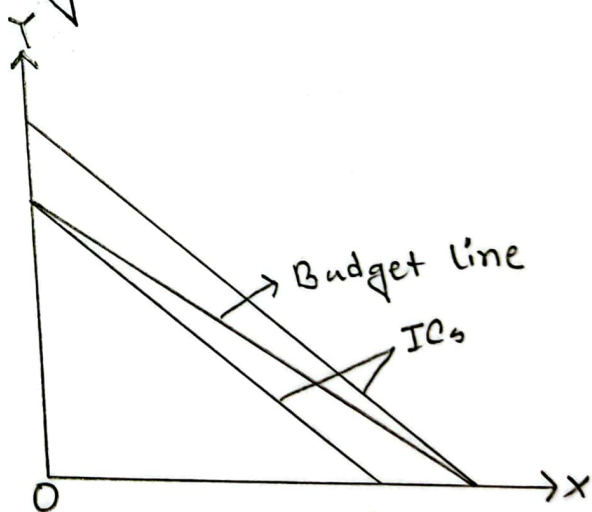


Fig: Flatter Budget line & Corner solution.

So, for maximum utility one should buy only X.

$$\text{Thus, } X^* = \frac{I}{P_x} = \frac{50}{2} = 25 \text{ \& } Y^* = 0$$

$\therefore$ Marshallian demands are $(X^*, Y^*) = (25, 0)$.

4. $U = \min(2x, Y)$. Find the demands for x & Y .

Answer:

Here, $U = \min(2x, Y)$ is a utility function for perfect complements.

There will be no MRS and tangency will be always at the corner i.e. where $2x = Y$ as shown in figure.

$$\begin{aligned} \text{Now, } MU_x &= 2 \text{ if } 2x < Y \\ &= 0 \text{ if } 2x \geq Y \end{aligned}$$

$$\text{Here, } I = P_x X + P_Y Y$$

$$\text{or, } I = P_x X + P_Y \cdot 2x$$

[Putting value of Y]

$$\text{or, } I = P_x X + 2P_Y X$$

$$\text{or, } I = X(P_x + 2P_Y)$$

$$\therefore X^* = \frac{I}{P_x + 2P_Y}$$

$$\therefore Y^* = 2\left(\frac{I}{P_x + 2P_Y}\right) \quad \text{or, } Y^* = \frac{2I}{P_x + 2P_Y}$$

$$\therefore \text{Marshallian demands are } (X^*, Y^*) = \left(\frac{I}{P_x + 2P_Y}, \frac{2I}{P_x + 2P_Y}\right)$$

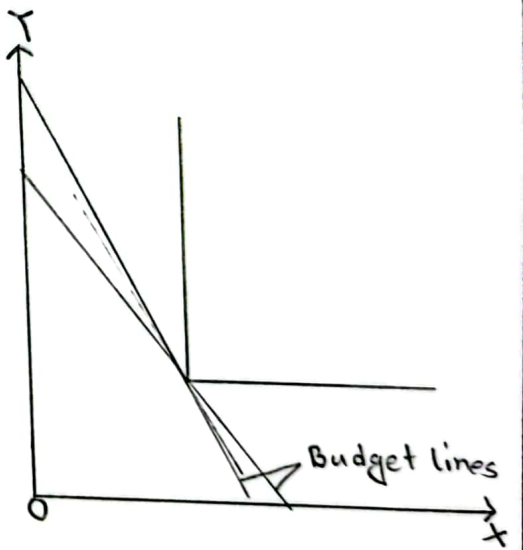


Fig: Demands for Perfect Complements

5. $U = X + \ln(Y)$. Find the demands for X & Y .

Answer:

$$\text{Here, Max}_{X,Y} U = X + \ln(Y)$$

$$\text{s.t. } I = P_X X + P_Y Y$$

$$\text{So, Max}_{X,Y,\lambda} L = X + \ln(Y) + \lambda [I - P_X X - P_Y Y]$$

$$\frac{\partial L}{\partial X} = 1 - \lambda P_X = 0 \quad \text{or, } 1 = \lambda P_X \quad \text{--- ①}$$

$$\frac{\partial L}{\partial Y} = \frac{1}{Y} - \lambda P_Y = 0 \quad \text{or, } \frac{1}{Y} = \lambda P_Y \quad \text{--- ②}$$

$$\frac{\partial L}{\partial \lambda} = I - P_X X - P_Y Y = 0 \quad \text{or, } I = P_X X + P_Y Y \quad \text{--- ③}$$

Dividing ① by ②,

$$\frac{1}{1/Y} = \frac{\lambda P_X}{\lambda P_Y}$$

$$\therefore Y^* = \frac{P_X}{P_Y}$$

Putting value of Y in ③,

$$I = P_x X + P_y \cdot \frac{P_x}{P_y}$$

$$\text{or, } I = P_x X + P_x$$

$$\text{or, } P_x X = I - P_x$$

$$\therefore X^* = \frac{I - P_x}{P_x}$$

Thus, Marshallian demands are -

$$X^* = \frac{I - P_x}{P_x} \quad \text{for } I \geq P_x$$

$$Y^* = \frac{P_x}{P_y} \quad \text{for } P_y > 0$$

These X^* & Y^* are connected for $I \geq P_x$

If $I < P_x$, then $X^* = 0$ and $Y^* = \frac{I}{P_y}$

6. For each of the following utility functions, check for diminishing MRS.

$$\text{a. } U = XY$$

Here, $U = XY$

$$\therefore MRS = \frac{U_x}{U_y} = \frac{Y}{X}$$

Now,

Using formula to check diminishing MRS.

$$\frac{dMRS}{dX} = \frac{\partial MRS}{\partial X} - MRS \cdot \frac{\partial MRS}{\partial Y}$$

$$= -\frac{Y}{X^2} - \frac{Y}{X} \cdot \frac{1}{X}$$

$$= -\frac{Y}{X^2} - \frac{Y}{X^2}$$

$$= -\frac{2Y}{X^2} < 0 \quad [\text{since } X \& Y \text{ both are goods}]$$

Thus, according to this formula, this utility function has diminishing MRS.

$$\underline{b. U = \ln X + \ln Y}$$

Here, $U = \ln X + \ln Y$

$$MRS = \frac{U_x}{U_y} = \frac{1/x}{1/y} = \frac{y}{x}$$

$$\frac{\partial MRS}{\partial x} = -\frac{y}{x^2} \quad \& \quad \frac{\partial MRS}{\partial y} = \frac{1}{x}$$

Now, Using formula to check diminishing MRS.

$$\frac{dMRS}{dx} = \frac{\partial MRS}{\partial x} - MRS \frac{\partial MRS}{\partial y}$$

$$= \frac{-y}{x^2} - \frac{y}{x} \cdot \frac{1}{x}$$

$$= -\frac{y}{x^2} - \frac{y}{x^2}$$

$$= -\frac{2y}{x^2} < 0 \quad [\text{since } x \& y \text{ are quantity of goods}]$$

Thus, according to this formula, this utility function has diminishing MRS.

$$\underline{C.U = X^2 + Y^2}$$

Here, $U = X^2 + Y^2$

$$\therefore MRS = \frac{U_x}{U_y} = \frac{2X}{2Y} = \frac{X}{Y}$$

$$\therefore \frac{\partial MRS}{\partial X} = \frac{1}{Y} \quad \& \quad \frac{\partial MRS}{\partial Y} = -\frac{X}{Y^2}$$

Now, using formula to check diminishing MRS-

$$\frac{dMRS}{dX} = \frac{\partial MRS}{\partial X} - MRS \frac{\partial MRS}{\partial Y}$$

$$= \frac{1}{Y} - \frac{X}{Y} \cdot \left(-\frac{X}{Y^2}\right)$$

$$= \frac{1}{Y} + \frac{X^2}{Y^3}$$

$$= \frac{1}{Y} \left(1 + \frac{X^2}{Y^2}\right) > 0 \quad \left[\text{since values of } x \& y \text{ are positive}\right]$$

Thus, according to this formula, this utility function does not have diminishing

MRS.

$$\underline{d. U = \ln X + Y}$$

Here, $U = \ln X + Y$

$$\therefore MRS = \frac{U_x}{U_y} = \frac{1/x}{1} = \frac{1}{x}$$

$$\frac{\partial MRS}{\partial X} = -\frac{1}{x^2} \quad \& \quad \frac{\partial MRS}{\partial Y} = 0$$

Now, using formula to check diminishing MRS.

$$\frac{dMRS}{dX} = \frac{\partial MRS}{\partial X} - MRS \cdot \frac{\partial MRS}{\partial Y}$$

$$= -\frac{1}{x^2} - \frac{1}{x} \cdot 0$$

$$= -\frac{1}{x^2} < 0 \quad [\text{since } x^2 \text{ is positive}]$$

Thus, according to this formula, this utility function has diminishing MRS.

e. $U = \min(x, y)$

$U = \min(x, y)$ is utility function for perfect complements. In this case, coefficients of x & y both are 1 and utility is maximized when $x = y$.

In case of this type of utility function, we cannot get MRS. A tangency will always be at the corner where $x = y$.

Thus, for $U = \min(x, y)$, we cannot check diminishing MRS.